

## Corsair's latest launch

Corsair launched P3 128 GB SSD which claims 400 MB/s read and 280 MB/s write speeds and supporting SATA III for a retail price of ₹14,000

## Kingston's HyperX SSDs

Kingston is expected to launch their HyperX line of SSDs housing the SandForce SF-2281 controller in capacities of 120GB and 240GB

Hard Drives Test



replaces traditional 512-byte hard drive sec-



With a 10,000 RPM spindle speed, it was difficult for other drives to beat the WD Velociraptor 600GB which fetched it the Best Performer

tors with 4 Kilobyte sectors, in its WD10EARS 3 TB drive whereas Seagate and Hitachi have used the traditional 512-byte sector for their 3 TB drives. These drives overcome the 2.19 TB limit (read more on this in the World View section of the January 2011 issue) by using a software (Seagate and Hitachi) or by using a host bus adaptor (in WD).

We got around five drives from WD, three from Seagate and one from Hitachi. We wonder why WD sent us the same models as last year: Caviar Black 1 TB and Caviar Green 1 TB. The only logical explanation we see is sales. Also the prices have dropped significantly making performance drives within reach.

WD Velociraptor WD6000HLHX (600GB) is double the capacity from its last iteration and has three 200 GB platters spinning at 10,000 RPM. It is SATA 6 Gbps compatible but do not expect a drastic jump in the real world scenario. In terms of performance, the Velociraptor pretty much ruled them all, but we found the data transfer scores

of the 3TB drives to be quite impressive. All of them use a 64MB of cache memory as opposed to 32 MB used in the Velociraptor. But more on them later.

WD WD1001FALS (1TB) is your Caviar Black edition performance drive and a very old one at that. Its performance was found to be slower than Seagate ST31000528AS (1TB) despite the same rotational speed and cache buffer. The difference between these two is the platters, where the Caviar Black 1 TB uses three platters, the Seagate one is more dense at 2 x 500GB platters.

Among the low power drives we got WD10EADS and WD10EARS (both 1TB) along with Seagate ST2000DL003 (2 TB). The only difference between the two WD drives is the cache buffer which gives the WD10EARS a slight edge over the other drive. Also the Caviar Green drives use an algorithm to determine the spindle speed which can be anywhere between 5400 - 7200 RPM. The Seagate 2 TB drive has a fixed rotational speed of 5900 RPM and we notice that this drive was faster than the other two green edition drives.



WD7500BPKT won the Best Buy for its great price to performance ratio among the Laptop HDDs

## HOW WE TESTED

We decided to separate the hard drives based on their categories such as Internal HDDs, Laptop HDDs and SSDs. We did not demarcate the drives within these categories based on capacity and technology used as ultimately a user will go for a drive based on his or her needs. Higher performance drives will tend to be priced higher than the large capacity value drives.

While testing hard drives and SSDs we have to make sure that there is no bottlenecking anywhere. To ensure this, we used two 600 GB Velociraptors in RAID 0 mode, which gave us excellent write speeds, so that it could take on even the fastest SSD to be tested. Our test rig is as shown in the box:

### TEST RIG

**Processor:** Intel Core i7-980X

**Motherboard:** ASUS Rampage III Black Edition

**RAM:** 2x2GB Corsair Dominator 1333MHz

**Operating System:** Windows 7 Ultimate

**OS Drive:** Intel SSDSA2SH080G1GN (80 GB)

**RAID 0 combination:** 2 x 600GB WD Velociraptor (SATA 3, 32 MB cache)

**Graphics Card:** Galaxy GTX 460

**Monitor:** BenQ E2200HD

**Power Supply:** Corsair HX 620W

Clean installs were done on each of the platforms and two partitions were created on the RAID 0 array. All extra Windows functions that can potentially slow down access, such as System Restore, File Indexing and a few unused services were disabled. The synthetic benchmarks were installed on the Intel SSD where as the real life file transfer benchmarks were installed on the primary RAID 0 partition. We used HD Tune Pro 4.0.1 for testing the Average Read / Write and Read Burst speeds as well as SiSoft Sandra 2011. Both these benchmarks require unformatted drives to perform write testing. After completing these tests, we formatted the test drives and created two partitions of equal size. For our Real world tests we used an 8 GB RAR file to simulate sequential operation to the disk and a folder of size 8 GB having various types of files of varying sizes to simulate a random or non sequential operation. These files were copied to and from the RAID 0 array to the drive under tests. We also performed these tests intra-drive which involves copying from one partition to the other except for SSDs. Finally we used Adobe Photoshop CS2 to test the amount of time taken to load a 1 GB image. Both the image and the scratch disk were kept on test drives on the same partition, effectively stimulating file access.

As hard drives are all about performance, we did not have any weightage for features, as there are none so to speak. Ultimately everything boils down to transfer speeds and pricing.

Finally coming to the 3TB warriors. As Seagate and Hitachi use the legacy 512-byte sector, it is comparatively easier for you to use these drives even if your OS is Windows XP and

in order to utilise the complete 3 TB all you have to do is install their respective drivers. With the WD 3 TB drive you will not be able to access the full capacity if you are using Win-